# Firm Expectations, Innovation and Growth

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October 13, 2026

# Motivation

- **Modern growth theory emphasizes the role of firms' R&D and innovation activity**

  (Klette and Kortum, 2004; Lentz and Mortensen, 2008; Akcigit and Kerr, 2018; Acemoglu et al., 2018)

- **Common assumption: Firms have Full-Information Rational Expectations**

- **Recent empirical research documents sizable deviations from this benchmark**
  - Systematic differences between firms' forecasts and realizations of production, sales, ...

    (Altig et al., 2022; Bachmann et al., 2021, Born et al. 2023, 2024)

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**We ask:**

What is the role of imperfect information on firm selection, innovation dynamics and aggregate productivity growth?

# This Paper

- **Empirical**
  - Panel of German firms with info on employment expectations and realizations
  - Firms underpredict employment growth on average
  - Substantial heterogeneity, esp. regarding firm age and firm exit/survival
  - Growth expectations are related to R&D activity

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- **Quantitative model**
  - Endogenous growth model with heterogeneous firms
  - Depart from **FIRE**
  - Calibrate the model to German economy
  - Study selection channels at work
  - Quantify the role of imperfect information on innovation and aggregate growth

# Literature and Contributions

- **Imperfect information, firm learning and ... selection** (Jovanovic (1982), Arkolakis, Papageorgiou, Timoshenko (2018)), **... export dynamics** (Albornoz, Pardo, Corcos, and Ornelas (2012), Eaton, Eslava, Jinkins, and Tybout (2021)), **... factor misallocation** (David, Hopenhayn, and Venkateswaran (2016))

  → Analyze the effects on firms' innovation activities

- **Quantitative analysis of firms' expectations using GE firm dynamics models** (Barrero (2022), Bachmann and Elstner (2015), Chen, Senga, Sun, and Zhang (2023))

  → Study the effects on aggregate productivity growth

- **Empirical analysis of firms' expectations and forecast errors** (Guiso and Parigi (1999), Bachmann, Elstner, and Sims (2013), Bachmann, Carstensen, Lautenbacher, and Schneider (2021), Altig, Barrero, Bloom, Davis, Meyer, and Parker (2022), Bloom, Davis, Foster, Lucking, Ohlmacher, and Saporta-Eksten (2020), Gennaioli, Ma, Shleifer (2016), Barrero (2022), Bachmann and Elstner (2015))

  → Employment forecast errors and their relationships with firm survival and R&D

# Roadmap

- **Empirical facts**
- **Model**
- **Quantitative analysis**

# Data

- **IAB Establishment Panel (1997-2019):** Annual, representative survey of German establishments
  - Information on various establishment characteristics: sales, intermediate inputs, workforce, investment, R&D, ...
  - $\sim$15,500 establishments each year $\rightarrow \sim$ 165,000 observations
- **Questions eliciting employment forecasts**

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- **Questions eliciting employment forecasts**

  Q1. *"Do you expect the total number of employees to remain constant by June of <Year>, or do you think it will increase or decrease?"*

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- **Questions eliciting employment forecasts**

  Q1. *"Do you expect the total number of employees to remain constant by June of <Year>, or do you think it will increase or decrease?"*

  Q2. *"What is the approximate expected total number of employees?"*

# Forecast Errors

**Use panel structure to derive realized and expected employment growth of establishment $i$**

Realized: $$g_{i,t+1} = \frac{n_{i,t+1} - n_{i,t}}{\frac{1}{2}(n_{i,t} + n_{i,t+1})}$$

Expected: $$g^{e}_{i,t+1} = \frac{E_t(n_{i,t+1}) - n_{i,t}}{\frac{1}{2}(n_{i,t} + E_t(n_{i,t+1}))}$$

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$\varepsilon_i > 0 \Rightarrow$ Underpredicting

$\varepsilon_i < 0 \Rightarrow$ Overpredicting

# Forecast Errors

| | **Employment growth** (in percent) | | **Forecast error** ($\epsilon = g - g^e$) |
|---|---|---|---|
| | **Expected** ($g^e$) | **Realized** ($g$) | (in pp) |
| **Mean** | **0.93** | **3.07** | **2.14** |
| **Std. dev.** | 17.34 | 25.25 | 27.17 |

On average, firms underpredict the growth of their workforce (with large dispersion)

Distribution Over time

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Distribution Over time

- **Relation with firm age?**
- **Do forecast errors differ between exiting and surviving firms?**
- **How do growth expectations relate to R&D activity?**

# Firm Age

| | Forecast error $\epsilon$ | Absolute value $\lvert\epsilon\rvert$ |
|---|---|---|
| **Age** (Reference: 0-2 years) | | |
| 3-5 | | $-0.028^{***}$ |
| 6-10 years | | $-0.049^{***}$ |
| 10+ years | | $-0.089^{***}$ |

Controls: Industry, firm size, region, year fixed effects.

- **Absolute forecast error decreases with age**

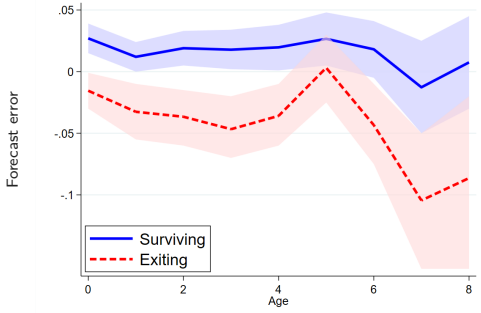
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| 6-10 years | $-0.015^{***}$ | $-0.049^{***}$ |
| 10+ years | $-0.008^{*}$ | $-0.089^{***}$ |

Controls: Industry, firm size, region, year fixed effects.

- **Absolute forecast error decreases with age**
- **Younger firms underpredict more**

# Firm Exit and Survival

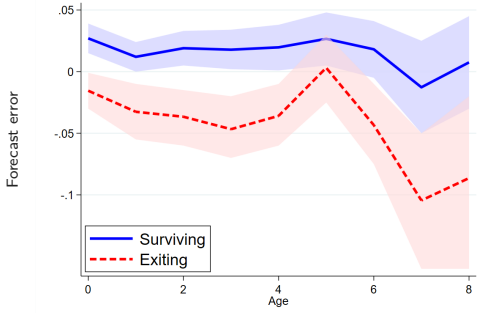



$$\varepsilon_{i,t+1} = \alpha + \beta_a \text{Age}_{i,t} + \beta_s \text{Survivor}_i + \beta_d \left(\text{Age}_{i,t} \times \text{Survivor}_i\right) + X_{i,t} + \gamma_t + \nu_{i,t} \ ,$$

$$\text{Survivor}_i = \text{No exit within first 10 years}$$

- **Firms exiting (within 10 years) overpredict employment growth**

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- **Firms exiting (within 10 years) overpredict employment growth**
- **Survivors (initially) underpredict**

# Growth Expectations and R&D

| | (1) **Innovation success ($t+1$)** | (2) **Deal with R&D ($t$)** | (3) **R&D intensity ($t$)** |
|---|---|---|---|
| **Expected growth ($t$)** (*reference*: $g^e = 0$) | | | |
| Negative | 0.017 | 0.020 | 0.004 |
| 0-22% | 0.123*** | 0.070*** | 0.014*** |
| Above 22% | 0.159*** | 0.087*** | 0.030*** |

Controls: Industry, firm size, region, year fixed effects, last realized growth

**Firms expecting to grow conduct more R&D and innovate more**

# Towards a Model

- **Recap**
  - Forecast errors decline with firm age
  - Surviving (exiting) firms underpredict (overpredict)
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- **Develop model that is consistent with these facts and use it for quantitative analysis**

- **Framework**
  Synthesis of $\underbrace{\text{Hopenhayn (1992)}}_{\text{Heterogeneous firms}}$ & $\underbrace{\text{Jovanovic (1986)}}_{\text{learn their productivity}}$ & $\underbrace{\text{Klette and Kortum (2004)}}_{\text{and innovate} \rightarrow \text{R\&D driven growth}}$

# Household

- **Infinitely-lived representative household**
- **Preferences**

$$\max \sum_{t=0}^{\infty} \beta^t \log(C_t) \qquad C_t = \left( \int_0^{M_t} c_{jt}^{\frac{\eta-1}{\eta}} dj \right)^{\frac{\eta}{\eta-1}}$$

  - $C_t$: Composite of consumption varieties
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- **Inelastic supply of one unit of labor**, earns wage $W_t$
- **Household holds a balanced portfolio of all firms**, $K_t$, earning the return $i_t$
- **Euler equation and demand function** (normalize $P_t \equiv \left( \int_0^{M_t} p_{jt}^{1-\eta} dj \right)^{\frac{1}{1-\eta}} = 1$)

$$\frac{C_{t+1}}{C_t} = \beta(1 + i_{t+1}) \qquad c_{jt} = (p_{jt})^{-\eta} C_t$$

# Firms: Production

- **Each consumption variety $j$ is produced by one firm**
- **Production technology**

$$y_t = exp(s_t) \cdot n_t$$

- **Firm's idiosyncratic productivity $s_t$**

$$s_t = z_t + \epsilon_t \qquad \epsilon_t \sim iid\ \mathcal{N}(0, \sigma_\epsilon^2)$$

  - $\epsilon_t$: Transitory productivity disturbance
  - $z_t$: Persistent productivity (next slide)

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- **Information friction: Firm cannot observe $z_t$ and $\epsilon_t$ separately, but only $s_t$**

# Firms: R&D and Innovation

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- **$z$ evolves stochastically according to**

$$z_{t+1} = \begin{cases} z_t + \zeta_{t+1} & \text{no innovation} \\ z_t + \zeta_{t+1} + \log(1+\lambda) & \text{innovation} \end{cases}$$

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- **Firm's productivity $s_t = z_t + \varepsilon_t$ changes over time due to ...**
  - ... innovation efforts
  - ... transitory shocks, $\epsilon_t$
  - ... persistent shocks, $\zeta_t$

# Firms: Beliefs, Learning and Choices

- **Firm forms belief $\hat{z}_t$ about productivity $z_t$**
- **Updating of beliefs:**

  Firm enters period $t$ with prior $\hat{z}_{t-1}$,
  observes $s_t$ and if innovation was successful,
  updates belief according to Kalman filter recursion:

$$\hat{z}_t = \begin{cases} \hat{z}_{t-1} + \mathcal{K}_t\big(s_t - \hat{z}_{t-1}\big) & \text{no innovation} \\ \hat{z}_{t-1} + \log(1+\lambda) + \mathcal{K}_t\big(s_t - \hat{z}_{t-1} - \log(1+\lambda)\big) & \text{innovation} \end{cases}$$

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  ($\mathcal{K}_t$: Kalman gain)

- **Choices in period $t$:** Exit/stay, production and R&D labor $n_t$ and $r_t$, output price $p_t$

# Value Function

$$V_t(\mathcal{S}_t) = \max_{\{\text{Exit, Stay}\}} \left\{ 0, \quad \max_{p_t, n_t, r_t} \left[ \pi_t + \frac{1-\delta}{1+i_t} \mathbb{E}_t V_{t+1}(\mathcal{S}_{t+1} | \mathcal{S}_t) \right] \right\}$$

- **Firm's state:** $\mathcal{S}_t = (s_t, \hat{z}_t)$
- **Choices**
  1. $\{\text{Exit, Stay}\}$
  2. $p_t$: Price, $n_t$: Labor, $r_t$: R&D
- **Profits** $\pi_t = p_t a_t n_t - W_t n_t - W_t r_t - W_t \gamma_f$

  with $a_t = exp(s_t)$ and $p_t = (a_t n_t / C_t)^{-1/\eta}$
- **Exogenous exit** with probability $\delta$
- **Firm's subjective expectation**: $\mathbb{E}_t$

# Firm Choices

- **Pricing rule under monopolistic competition**

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  - More productive firms have higher employment

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  - More productive firms have higher employment

- **Innovation probability** $x(r_t)$

$$W_t = x'(r_t) \frac{1 - \delta}{1 + i} \mathbb{E}_t \Big[ V_{t+1} \left( \text{innovation} \right) - V_{t+1} \left( \text{no innovation} \right) \Big]$$

  - Higher productivity belief $\hat{z}_t$ leads to higher perceived marginal benefit of R&D $\Rightarrow$ $r_t$ and $x(r_t)$ increasing in $\hat{z}_t$

# Entry and Equilibrium

- **Entrants**
  - Pay fixed entry cost $\gamma_e \geq 0$ (in units of labor)
  - Draw initial productivity $z_1$ from a distribution whose mean grows with aggregate productivity
  - Observe $s_1 = z_1 + \epsilon_1 \;\rightarrow\;$ Beliefs $\hat{z}_1 \;\rightarrow\; \{\text{Stay, Exit}\}$

- **Free entry condition**

- **Balanced growth path equilibrium**
  - Aggregate variables grow at constant rate
  - Stationary distribution of firms

# Quantitative Analysis (in progress)

- **Calibration**
  - Set parameters to match data targets from IAB-EP and aggregate statistics
  - Fit of non-targeted moments
- **Study main mechanisms at work**
  - Two important selection channels
- **Counterfactual experiment**
  - Quantify role of imperfect information

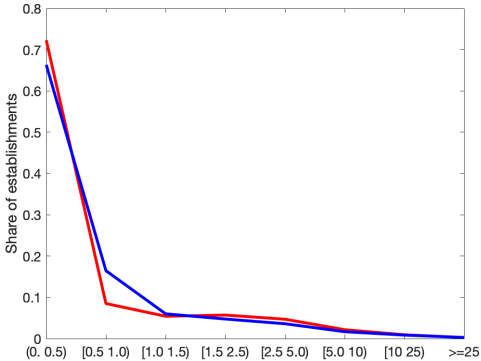
# Parameters

| Symbol | Value | Description | Source |
|---|---|---|---|
| **Household** | | | |
| $\beta$ | 0.976 | Discount factor | $\beta = \frac{1+g}{1+i}$ |
| $\eta$ | 3.0 | Elasticity of substitution | Broda Weinstein (2006) |
| **Firm shocks** | | | |
| $\delta$ | 0.01521 | Exogenous firm exit | Calibrated |
| $\sigma_\epsilon^2$ | 0.00381 | Var. transitory shock | Calibrated |
| $\sigma_\zeta^2$ | 0.000495 | Var. persistent shock | Calibrated |
| **Distribution of initial productivity** $z_1$ | | | |
| $\mu_e$ | -0.505 | Mean | Calibrated |
| $\sigma_e^2$ | 0.5603 | Variance | Calibrated |
| **Fixed costs** | | | |
| $\gamma_f$ | 0.047 | Fixed cost of production | Calibrated |
| $\gamma_e$ | 2.223 | Fixed cost of entry | Calibrated |
| **Innovation and R&D technology** $r = \frac{\rho}{\psi} A^\varphi x^\psi$ | | | |
| $\lambda$ | 0.0193 | Productivity increase after innovation | Calibrated |
| $\rho$ | 0.36 | Scaling factor | Calibrated |
| $\varphi$ | 1.28 | Elasticity wrt firm productivity | Calibrated |
| $\psi$ | 2.0 | Elasticity wrt innovation probability | Acemoglu et al. (2013), Akcigit and Kerr (2018) |

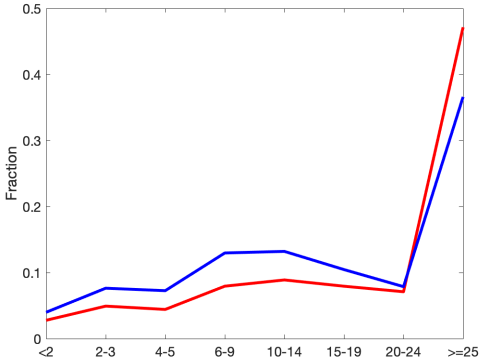
# Internally Calibrated Parameters: Model and Data

| Description | Model | Data | Parameter |
|---|---|---|---|
| Growth rate of aggregate productivity | 1.5% | 1.5% | $\lambda$ |
| Coefficient of variation of employment | 6.09 | 6.09 | $\sigma_{\zeta}^2$ |
| Coefficient of variation of employment for young firms | 2.88 | 2.88 | $\sigma_{e}^2$ |
| Exit rate of young establishments (age 0-2 years) | 7.3% | 7.3% | $\gamma_f$ |
| Exit rate of establishments | 2.8% | 2.8% | $\delta$ |
| Mean employment of young firms | 1.0 | 1.0 | $\mu_e$ |
| Forecast error of employment growth | 2.0% | 2.0% | $\sigma_{\epsilon}^2$ |
| R&D investment to GDP ratio | 0.028 | 0.028 | $\rho$ |

# Model and Data

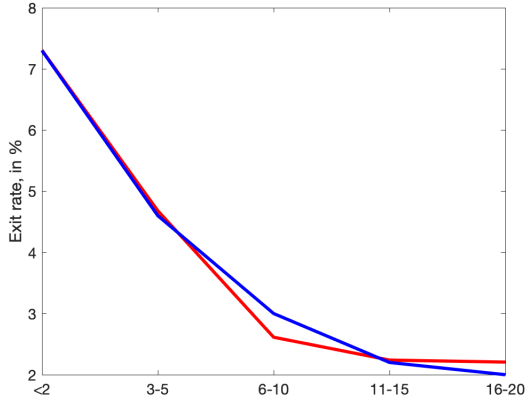



Size distribution



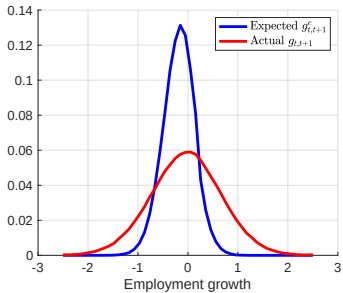

Age distribution

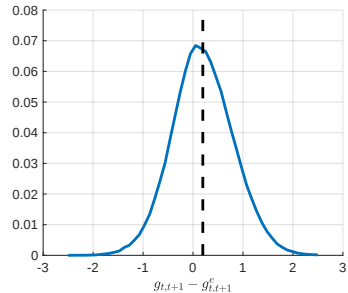
# Model and Data




Exit rate by age

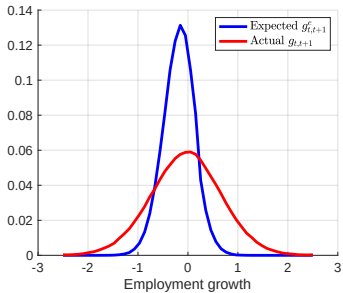
# Employment Growth and Forecast Errors

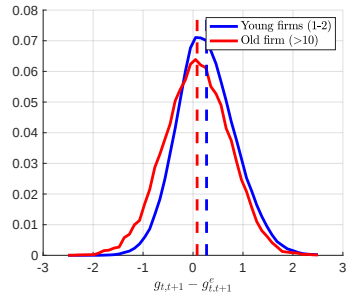



- **Actual employment growth more dispersed than expected employment growth**

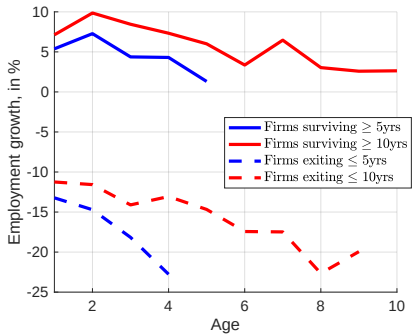
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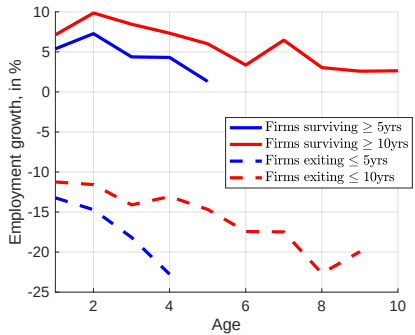

- **Actual employment growth more dispersed than expected employment growth**
- **Larger forecast errors for young firms**

# Selection Effects at Work




- **Heterogeneous firm dynamics is governed two types of selection effects**
  1. Classical selection effect (Hopenhayn 1992) governed by productivity shocks

# Selection Effects at Work




- **Heterogeneous firm dynamics is governed two types of selection effects**
  1. Classical selection effect (Hopenhayn 1992) governed by productivity shocks
  2. Selection effect due to heterogenous beliefs

# Selection Due to Heterogenous Beliefs

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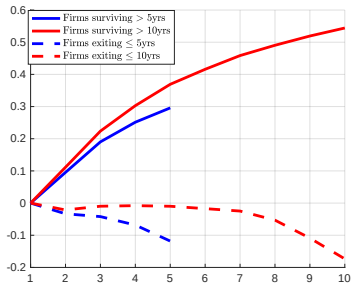
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$\Rightarrow$ **Initially underpredicting firm:** Positive employment growth and more likely to survive

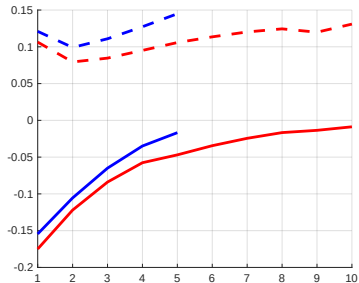
**Initially overpredicting firm:** Negative employment growth and more likely to exit

# Selection Due to Heterogenous Beliefs

- **Surviving firms underpredict initially and revise beliefs upwards**
- **Exiting firms overpredict initially and over time (until exit)**




(a) Change in beliefs $\hat{z} - \hat{z}_1$




(b) Overprediction $\hat{z} - z$

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  $\Rightarrow$ If learning is sufficiently slow, then **(initial) beliefs become self-fulfilling**

# Counterfactual

- **Eliminate information friction: Firm can observe $z$**

| | $g$ (%) | $\frac{R\&D}{GDP}$ (%) | Entry rate (%) | Exit rate (%) 1yr | 2yrs | 5 yrs |
|---|---|---|---|---|---|---|
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  - $\rightarrow$ Higher aggregate productivity growth

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| **FIRE** | **2.1** | **3.6** | 2.5 | 7.7 | 5.9 | 3.3 |

$\Rightarrow$ **Imperfect information lowers aggregate growth by $\sim$0.6 pp**

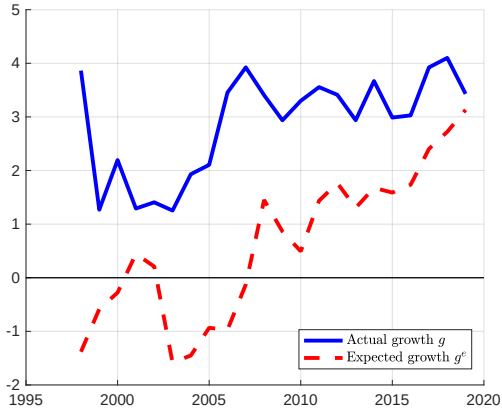
# Conclusion

- **We analyze subjective employment expectations in a panel of German firms**
  - Sizable forecast errors which are larger for younger firms
  - They relate to firm survival and R&D activity

- **We quantitatively assess the role imperfect information for innovation and growth**
  - Important role of imperfect information for firm selection
  - Under-predicting growth of (ultimately surviving) firms can depress R&D and aggregate growth

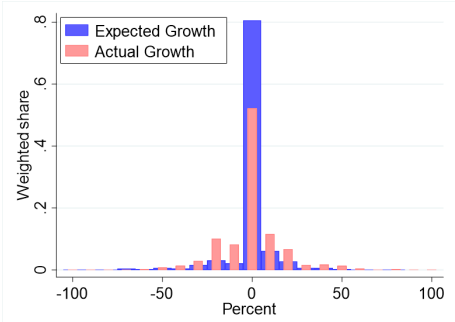
- **Next steps**
  - Decomposition of total growth effect into reallocation, selection and net entry
  - Welfare analysis
  - Policy: Targeted R&D subsidies

# Evolution Over Time

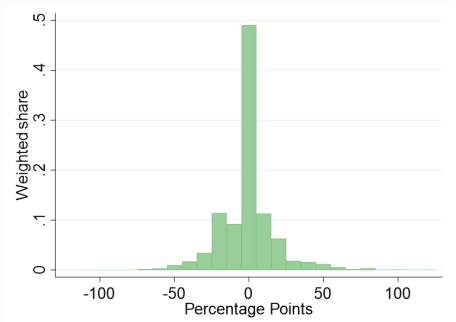



- **Firms consistently underpredict their employment growth during 1997-2019**
- **They increasingly underpredict job destruction $\Rightarrow$ Declining forecast errors**

# Distribution of Employment Growth, Expectations and Forecast Errors




(a) Expected/actual employment growth




(b) Forecast error

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# Initial Beliefs Matter for Exit

| **Exit rate** | **1 year** | **2 years** | **3 years** | **5 years** |
|---|---|---|---|---|
| Over-predictor | 0.034 | 0.054 | 0.074 | 0.161 |
| | (0.009) | (0.012) | (0.015) | (0.027) |
| Under-predictor | 0.025 | 0.040 | 0.071 | 0.127 |
| | (0.008) | (0.010) | (0.016) | (0.023) |

Initial belief = Forecast error at age one

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# Forecast Errors Predict Subsequent Exit

| **Exit between $t+1$ and $t+x$** | **1 year** | **2 years** | **3 years** | **5 years** |
|---|---|---|---|---|
| Forecast error | $-0.023^{***}$ | $-0.026^{***}$ | $-0.035^{***}$ | $-0.046^{***}$ |
| | (0.004) | (0.005) | (0.007) | (0.011) |

Linear probability model

$$P(Exit_{i,[t+1,t+x]}) = \alpha + \beta\epsilon_{i,t,t+1} + \gamma_t + X_{i,t} + \nu_{i,t}$$

(firm control $X_{i,t}$, include size, industry, and state)

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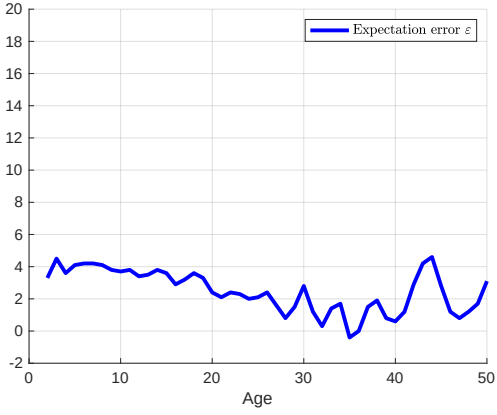
# Learning

$$\ln(E_{t+1}[n_{i,t+2}]) - \ln(E_t[n_{i,t+1}]) = \alpha + \beta e_{i,t,t+1} + \Gamma\left(e_{i,t,t+1} \times \text{Age}_{i,t}\right) + \gamma_t + \nu_{i,t}.$$

| | **Belief revision** | | |
|---|---|---|---|
| Forecast error | $0.882^{***}$ | $0.968^{***}$ | $0.950^{***}$ |
| × **Age** | | | |
| 1 - 2 years | | 0.059 | $-0.062$ |
| 3 - 5 years | | $-0.001$ | $-0.014$ |
| 6+ years | | $-0.449^{*}$ | $-0.418^{*}$ |
| Year FE | ✓ | ✓ | ✓ |
| Size | | | ✓ |

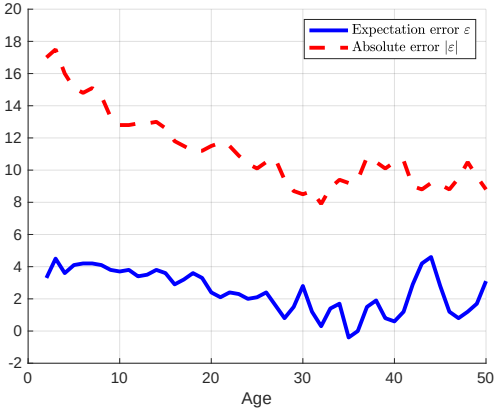
**Firms revise beliefs in response to forecast errors, and less so when they are older**

# Forecast error in balanced panel




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# Forecast error in balanced panel




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